Fire-aware robot dogs

Machine-learning wildfire forecasts for safe navigation and scouting

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Why

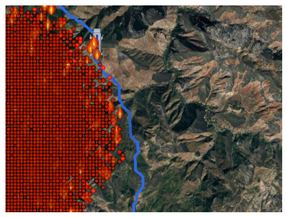
The edge of a wildfire is one of the most dangerous places for a firefighter.

Robot dogs (four-legged robots) can climb steep, rocky ground where trucks get stuck, and measure conditions on the ground under smoke and trees.

But a dog that steers by where the fire is now can walk into where it will be.

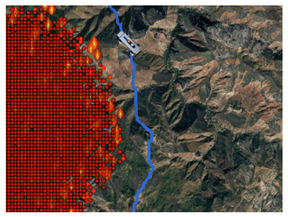
Plan on the predicted fire

Planned on the fire now



In the flames
75 min in, the fire has reached the route.

Planned on the prediction



Still clear
Same moment: this route keeps 100 m from the predicted fire.

The dog plans to stay at least 100 m from where the fire is predicted to be, with 30 min in hand.

It scouts from the closest spot that keeps that distance and still has a way home. When its sensors show the real fire differs, the prediction reruns and the dog re-plans. If it is cut off, an emergency retreat may come closer.

Simulator screenshots, Big Sur: same fire, same start and target, 75 minutes after the dog sets out.

How we predict the fire

Fire history + conditions
246,000 NASA satellite fire detections from 30 wildfires; weather, terrain, land cover

→

Machine learning + fire simulator
XGBoost gives each spot's chance of catching fire; the simulator spreads it with the wind

→

When each spot burns
Every 50 m spot, in 15 min steps

Small and fast, so the dog can re-plan on the spot.

31 KB
The whole model. Smaller than one photo on this poster.

0.1 s
One model pass over all 360,000 spots on the map (desktop CPU).

No GPU
No internet needed once loaded. Built for small edge computers; Raspberry Pi tests are next.

Model trained to predict whether a fire keeps spreading on 24 fires, tested on 6 others (AUC 0.75). The 50 m, 15 min spread is a simulation, not checked against real fires.

Close enough to help, safe enough to return

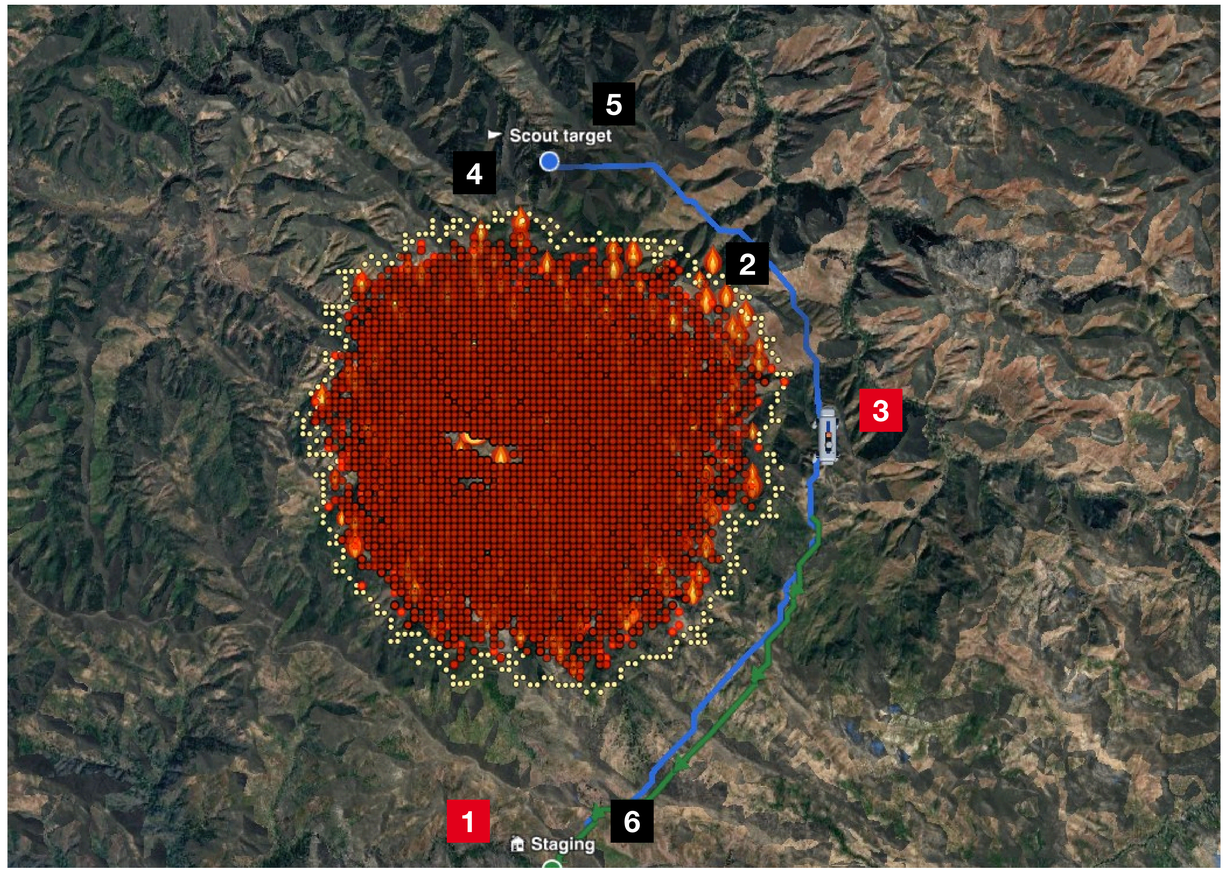
430 simulated missions per dog

	Reached a spot to scout	Caught by the fire
Reacts, keeps 500 m away	0%	0%
Reacts, keeps 200 m away	27%	0.2%
Predicts, no margin	100%	46%
Predicts, 100 m + 30 min + sensing	91%	0.5%

Reacting stays safe by staying back.

One afternoon at a Big Sur fire

A simulated mission on real terrain. Follow the dog from stop 1 to stop 6. How it decides at each stop: right panel.



1 **2:45 PM** Launch
Steep, brushy slopes, burning since midnight. Too dangerous for a crew, so the dog leaves staging.

2 **2:45 PM** Plan
The **blue** route is planned to keep 100 m from the forecast fire, with 30 min in hand, slower in brush and uphill.

3 **3:15 PM** Wind shift
The real fire turns; the forecast doesn't know, so the prediction needs a rerun. The **green** escape path stays open.

4 **4:45 PM** Scout
It reaches its scout spot, as close as the 100 m rule allows.

5 **4:45 PM** Measure
Its sensors find wind from the northwest, not the west, and 40 °C, not 35. The prediction reruns: the fire is coming, so the dog re-plans and heads home.

6 **~7:15 PM** Home safe
Back before the fire arrives. The crew plans with a better forecast.

Know where the fire will be, not just where it is.

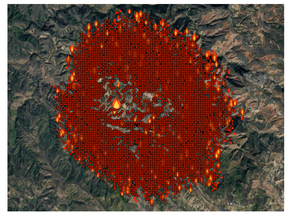
Safer crews
Fewer firefighters sent into danger to

Robot stays safe
Routes and exits lower the risk of

Smarter sensing
Measure where it changes decisions.

How the dog decides

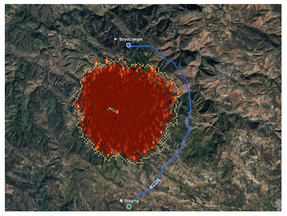
1 Predict



50 m cells **15 min** steps

Forecasts when each cell will burn.

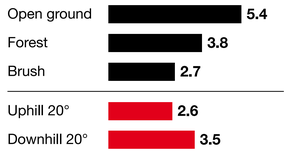
2 Safe route



Blue route to the scout target

Plans to keep 100 m from the forecast fire, with 30 min in hand.

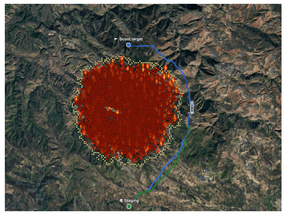
2b Route speed



Walking speed in **km/h**

The route is timed at these speeds. The planner never crosses water or slopes over 35°.

3 Escape path



Green escape path home

Aims to pass each point home an hour before the forecast fire comes within 100 m.

4 Scout



Wind, temperature, humidity, fire edge

Measures what the forecast only guessed.

5 Feedback



Launch **84%** Before report **78%** After **81%**

The readings improve the next forecast.

Simulator views on real terrain. Yellow dotted edge: the forecast the dog plans on; the real fire is hidden from it until its sensors see it. Forecast match: overlap of the forecast's and the real fire's new spread 4 h ahead, both simulated. Card numbers are planning steps, not the story's stops. Robot image: design concept; sensors are simulated.

Status and next

Done	Next
ML fire forecast trained and tested	Forecast + planner on the dog's controller
3D browser simulator on real terrain	Real sensors feeding the forecast
Containment planner (search)	Scout the planned containment line
Robot-dog loop in mock simulation	Tests on real robot dogs

Open-source code and demo video

Data and references

Code: github.com/sxz2012/wildfire-simulation

NASA FIRMS (VIIRS active-fire detections); Open-Meteo weather archive; ESA WorldCover 10 m land cover; Cesium World Terrain; Google Photorealistic 3D Tiles

Rothermel, R. C. (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service INT-115.

Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD '16.*